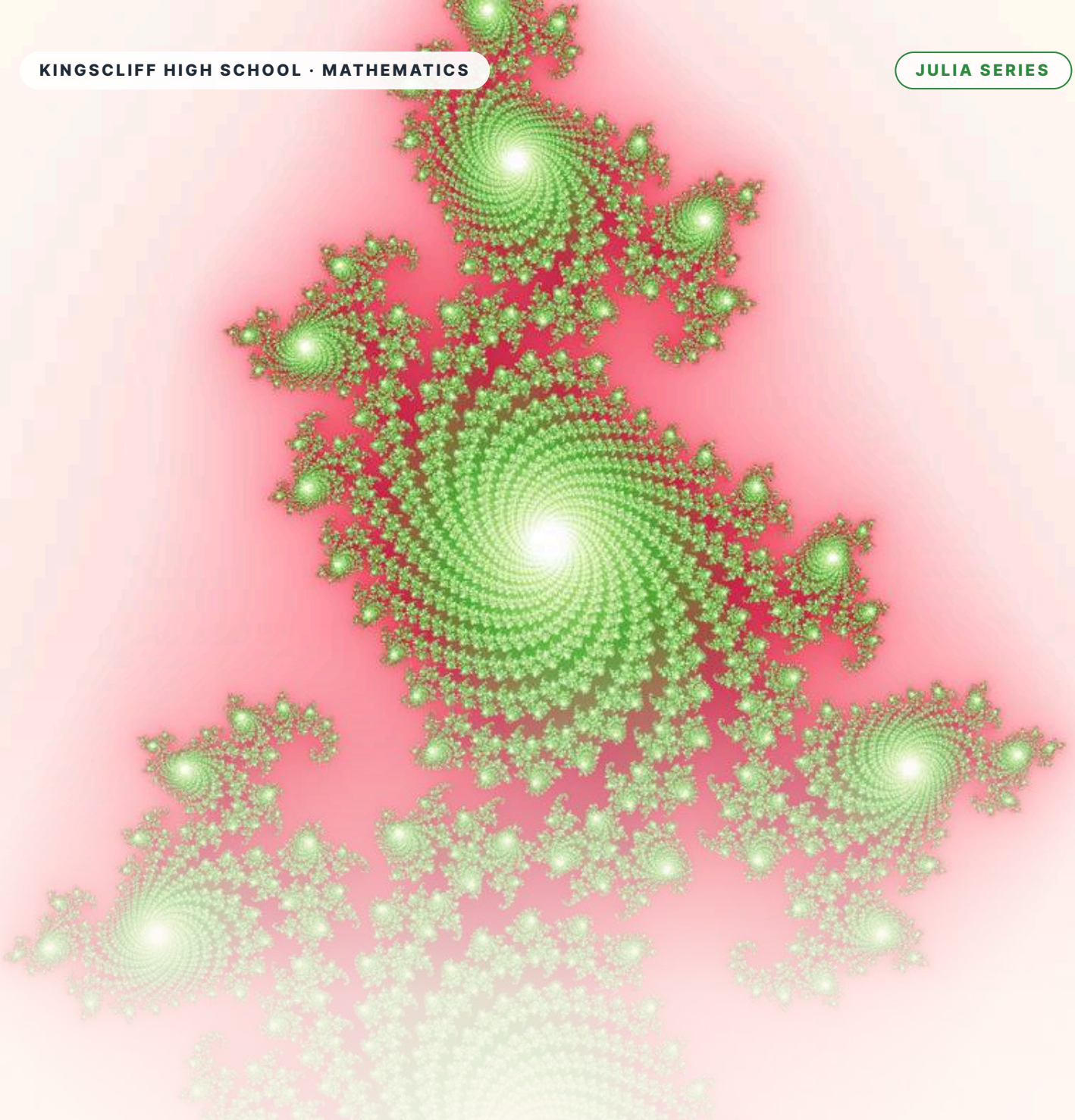
KINGSCLIFF HIGH SCHOOL · MATHEMATICS

JULIA SERIES



YEAR 7 → YEAR 8

Summer Maths

Ready for Year 8

Six weeks. Twenty questions a week. Every Year 7 skill that Year 8 builds on – mixed together so it sticks.

Week 1 | Week 2 | Week 3 | Week 4 | Week 5 | Week 6

NAME _____

CLASS _____

How it works

One page a week

Each week has **20 questions** and a **bonus**. Set aside about **40 minutes** – one sitting or a few short ones. Questions start easier and get a little harder through the page.

The same pattern every week

Question 1 is always integers, question 11 is always perimeter and circles, and so on. If the same number keeps going wrong, that is the skill to revise before Year 8.

1	Integers	11	Perimeter & circles
2	Fractions	12	Area
3	Decimals & percentages	13	Volume
4	Indices & primes	14	Number plane
5	Squares & roots	15	Patterns & rules
6	Simplify	16	Ratios & rates
7	Substitute	17	Angles
8	Expand & factorise	18	Probability
9	Equations	19	Data
10	Equations	20	Problem solving

Why Julia? Each banner is part of the same *Julia set*: pick a number c, then repeat $z \to z^2 + c$ for every point. Points that stay small are coloured in. Every booklet in the series zooms into a different corner of the same set – this one is the whole set, turned on its side.

NSW syllabus (NESA Mathematics K–10, 2022). Revises MA4-INT-C-01, MA4-FRC-C-01, MA4-IND-C-01, MA4-RAT-C-01, MA4-ALG-C-01, MA4-EQU-C-01, MA4-LIN-C-01, MA4-LEN-C-01, MA4-ARE-C-01, MA4-VOL-C-01, MA4-ANG-C-01, MA4-GEO-C-01, MA4-DAT-C-01, MA4-DAT-C-02, MA4-PRO-C-01. ★ Year 8 preview: MA4-PYT-C-01 and the Year 8 parts of MA4-IND, MA4-ALG, MA4-EQU, MA4-ARE and MA4-VOL.

Rules of the game

- **No calculator.** When a circle question says so, use $\pi \approx 3.14$.
- **Show working** in the space under each question.
- **Mark it yourself** using the answers at the back (pages 10–12). Each answer has a line of working to help you find a mistake.
- **Fix it.** Redo anything you got wrong a day or two later, without looking.
- ★ questions preview Year 8 work. Have a go – they are the ones that will help most in Term 1.
- **Stuck?** The hints page (page 9) has the key facts and formulas.

My summer tracker

WEEK	DATE	SCORE	DONE	BONUS	REVISE (Q NUMBERS)
1 Week 1		/ 20	☐	☐	
2 Week 2		/ 20	☐	☐	
3 Week 3		/ 20	☐	☐	
4 Week 4		/ 20	☐	☐	
5 Week 5		/ 20	☐	☐	
6 Week 6		/ 20	☐	☐	

First up in Year 8: Pythagoras' theorem, working with numbers (index laws), algebra (expanding and factorising), area of circles and trapeziums, and equations. Every one of them starts from the skills in these six pages.

WEEK 1

Summer Maths

Ready for Year 8

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Patterns & graphs · Angles & chance · Data & context

★ Year 8 preview (Stage 4) · No calculator

1 INTEGERS

Evaluate $-7 + 12$

2 FRACTIONS

Evaluate $\frac{2}{5} + \frac{1}{5}$

3 DECIMALS & PERCENTAGES

Write 0.35 as a percentage.

4 INDICES & PRIMES

Evaluate 2^4

5 SQUARES & ROOTS

Evaluate $\sqrt{49}$

6 SIMPLIFY

Simplify $5a + 3a - 2a$

7 SUBSTITUTE

If $m = 4$, find the value of $3m + 5$.

8 EXPAND & FACTORISE

Expand $3(x + 4)$

9 EQUATIONS

Solve $x + 9 = 15$

10 EQUATIONS

Solve $4x = 28$

11 PERIMETER & CIRCLES

Find the perimeter of a rectangle 9 cm long and 4 cm wide.

12 AREA

Find the area of a triangle with base 10 cm and perpendicular height 6 cm.

13 VOLUME

Find the volume of a rectangular prism 5 cm long, 4 cm wide and 2 cm high.

14 NUMBER PLANE

In which quadrant is the point $(-3,\ 2)$?

15 PATTERNS & RULES

Complete the table for $y = x + 3$.

x	0	1	2	3
y				

16 RATIOS & RATES

Simplify the ratio $12 : 18$

17 ANGLES

Find a.



18 PROBABILITY

A fair six-sided die is rolled. Find $P(\text{even number})$.

19 DATA

Find the mean of $4,\ 7,\ 9,\ 12,\ 8$.

20 PROBLEM SOLVING

At Thredbo it was $-4°$C at 6 am. By midday the temperature had risen 11 degrees. What was the midday temperature?

BONUS Find three consecutive whole numbers that add to 48.

WEEK 2

Summer Maths

Ready for Year 8

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Patterns & graphs · Angles & chance · Data & context

★ Year 8 preview (Stage 4) · No calculator

1 INTEGERS

Evaluate -3×6

2 FRACTIONS

Simplify $\frac{18}{24}$

3 DECIMALS & PERCENTAGES

Find 25% of \$60.

4 INDICES & PRIMES

Write 36 as a product of prime factors in index form.

5 SQUARES & ROOTS

Evaluate $5^2 - \sqrt{16}$

6 SIMPLIFY

Simplify $2a \times 5b$

7 SUBSTITUTE

If $a = 3$ and $b = 5$, find the value of $ab - a$.

8 EXPAND & FACTORISE

Expand $5(2y - 3)$

9 EQUATIONS

Solve $x - 8 = 3$

10 EQUATIONS

Solve $2x + 5 = 17$

11 PERIMETER & CIRCLES

Find the circumference of a circle with diameter 10 cm. Use $\pi \approx 3.14$.

12 AREA

Find the area of a parallelogram with base 8 cm and perpendicular height 5 cm.

13 VOLUME

A cube has edges 3 cm long. Find its volume.

14 NUMBER PLANE

Which of these points lies on the x-axis: $(0,\ 5)$, $(5,\ 0)$ or $(5,\ 5)$?

15 PATTERNS & RULES

Find the rule linking x and y.

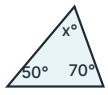
x	1	2	3	4
y	5	6	7	8

16 RATIOS & RATES

A car travels 240 km in 3 hours. Find its average speed.

17 ANGLES

Find x.



18 PROBABILITY

A bag holds 4 red and 6 blue counters. One is chosen at random. Find $P(\text{red})$.

19 DATA

Find the median of $3,\ 9,\ 4,\ 7,\ 5$.

20 PROBLEM SOLVING

A theme park charges \$85 per adult and \$65 per child. Find the cost for 2 adults and 3 children.

BONUS Two whole numbers have a product of 36 and a sum of 13. Find them.

WEEK 3

Summer Maths

Ready for Year 8

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Patterns & graphs · Angles & chance · Data & context

★ Year 8 preview (Stage 4) · No calculator

1 INTEGERS

Evaluate $-15 \div (-3) + 4$

2 FRACTIONS

Evaluate $\frac{2}{3} + \frac{1}{4}$

3 DECIMALS & PERCENTAGES

Evaluate 0.6×0.4

4 INDICES & PRIMES

Which of these are prime: 21, 23, 27, 29?

5 SQUARES & ROOTS

Between which two consecutive whole numbers does $\sqrt{30}$ lie?

6 SIMPLIFY

Simplify $7x + 4y - 3x + y$

7 SUBSTITUTE

If $p = -2$, find the value of $5 - 3p$.

8 EXPAND & FACTORISE ★

Factorise $6x + 15$

9 EQUATIONS

Solve $\frac{x}{4} = 7$

10 EQUATIONS

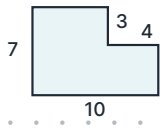
Solve $3x - 4 = 11$

11 PERIMETER & CIRCLES

A square has a perimeter of 36 cm. How long is each side?

12 AREA

Find the area. (cm)



13 VOLUME

A fish tank is 50 cm long, 30 cm wide and 20 cm high. Find its volume.

14 NUMBER PLANE

How far apart are the points $(2,\ 3)$ and $(2,\ -4)$?

15 PATTERNS & RULES

Complete the table for $y = 2x - 1$.

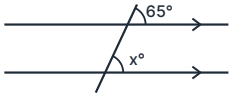
x	0	1	2	3
y				

16 RATIOS & RATES ★

Share \$45 in the ratio $2 : 3$.

17 ANGLES

The lines are parallel. Find x.



18 PROBABILITY

The chance of rain tomorrow is 0.3. What is the chance it does not rain?

19 DATA

Find the range and the mode of $6,\ 9,\ 4,\ 9,\ 12,\ 7$.

20 PROBLEM SOLVING

A 2 L bottle of juice costs \$3.60 and a 600 mL bottle costs \$1.20. Which is better value?

BONUS

Use each of the digits 1, 2, 3 and 4 once to make two 2-digit numbers with the largest possible product.

WEEK 4

Summer Maths

Ready for Year 8

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Patterns & graphs · Angles & chance · Data & context

★ Year 8 preview (Stage 4) · No calculator

1 INTEGERS

Evaluate $20 - 4 \times (-3)$

2 FRACTIONS

Evaluate $\frac{3}{4} \times \frac{8}{9}$

3 DECIMALS & PERCENTAGES

Write $\frac{3}{8}$ as a decimal.

4 INDICES & PRIMES

Write $5 \times 5 \times 5 \times 5$ in index form, then evaluate it.

5 SQUARES & ROOTS

Evaluate $\sqrt[3]{64} + \sqrt{81}$

6 SIMPLIFY

Simplify $12ab \div 4b$

7 SUBSTITUTE

If $x = 3$ and $y = -2$, find the value of $x^2 + y$.

8 EXPAND & FACTORISE ★

Factorise $8m - 12$

9 EQUATIONS

Solve $15 - x = 9$

10 EQUATIONS ★

Solve $5x + 3 = 2x + 18$

11 PERIMETER & CIRCLES

A rectangular paddock is 120 m by 80 m. Find its perimeter in kilometres.

12 AREA ★

Find the area of a trapezium with parallel sides 5 cm and 9 cm and perpendicular height 4 cm.

13 VOLUME

A triangular prism has a cross-section of area 15 cm² and length 8 cm. Find its volume.

14 NUMBER PLANE

$A(1,\ 1)$, $B(5,\ 1)$ and $C(5,\ 4)$ are three corners of rectangle $ABCD$. Find D.

15 PATTERNS & RULES

One square uses 4 matches, 2 squares in a row use 7 and 3 squares use 10. How many matches for 10 squares?

16 RATIOS & RATES

4 mangoes cost $9. How much would 10 mangoes cost?

17 ANGLES

Find x.



18 PROBABILITY

A letter is chosen at random from the word SUMMER. Find $P(\text{M})$.

19 DATA

Find the median of the data in the stem-and-leaf plot.

1	2 5 8
2	0 3 3 7
3	1 4

Key: 2 | 3 means 23

20 PROBLEM SOLVING ★

A canteen sold 120 pies on Monday. On Tuesday it sold 15% more. How many pies did it sell on Tuesday?

BONUS

A rectangle has a perimeter of 30 cm and its length is twice its width. Find its area.

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Patterns & graphs · Angles & chance · Data & context

★ Year 8 preview (Stage 4) · No calculator

1 INTEGERS

Evaluate $-6-(-10)$

2 FRACTIONS

Evaluate $1\frac{1}{2}+2\frac{2}{3}$

3 DECIMALS & PERCENTAGES

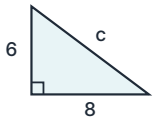
Find 15% of \$80.

4 INDICES & PRIMES ★

Write as a single power: $3^4 \times 3^5$

5 SQUARES & ROOTS ★

Find c. (cm)



6 SIMPLIFY

Simplify $3x \times 4x$

7 SUBSTITUTE

$P = 2l + 2w$. Find P when $l = 7.5$ and $w = 4$.

8 EXPAND & FACTORISE ★

Expand and simplify $2(x+5)+3(x-1)$

9 EQUATIONS

Solve $6x - 4 = 20$

10 EQUATIONS

Solve $\frac{x}{3}+4=10$

11 PERIMETER & CIRCLES

Find the circumference of a circle with radius 5 m. Use $\pi \approx 3.14$.

12 AREA ★

Find the area of a circle with radius 10 cm. Use $\pi \approx 3.14$.

13 VOLUME

A container holds 2500 cm³. How many litres is this?

14 NUMBER PLANE

Does the point $(3,\ 7)$ lie on the line $y = 2x + 1$?

15 PATTERNS & RULES

Find the rule linking x and y.

x	0	1	2	3
y	2	7	12	17

16 RATIOS & RATES

A recipe uses flour and sugar in the ratio $5:2$. How much sugar goes with 400 g of flour?

17 ANGLES

An isosceles triangle has an apex angle of $40°$. Find each base angle.

18 PROBABILITY

A spinner has 8 equal sections numbered 1 to 8. Find $P(\text{number greater than } 6)$.

19 DATA

The mean of four numbers is 9. Three of them are 6, 10 and 12. Find the fourth.

20 PROBLEM SOLVING

A pool 8 m long and 4 m wide has a path 1 m wide all the way around it. Find the area of the path.

BONUS What is the smallest whole number that 2, 3, 4, 5 and 6 all divide into exactly?

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Patterns & graphs · Angles & chance · Data & context

★ Year 8 preview (Stage 4) · No calculator

1 INTEGERS

Evaluate $(-4)^2 - 3 \times (-5)$

2 FRACTIONS

Evaluate $\frac{3}{5} \div \frac{9}{10}$

3 DECIMALS & PERCENTAGES

Order from smallest to largest: 0.45, $\frac{2}{5}$, 42%

4 INDICES & PRIMES ★

Write as a single power: $7^8 \div 7^3$

5 SQUARES & ROOTS ★

Is a triangle with sides 5 cm, 12 cm and 13 cm right-angled? Explain.

6 SIMPLIFY

Simplify $\frac{15x^2}{5x}$

7 SUBSTITUTE

If $a = -3$ and $b = 4$, find the value of $a^2 + 2b$.

8 EXPAND & FACTORISE ★

Factorise $5x^2 + 10x$

9 EQUATIONS ★

Solve $7x - 5 = 3x + 11$

10 EQUATIONS

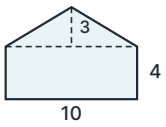
I think of a number, multiply it by 4 and subtract 7. The answer is 25. Write an equation and solve it.

11 PERIMETER & CIRCLES

A square has an area of 49 cm². Find its perimeter.

12 AREA

Find the area. (cm)



13 VOLUME ★

Find the volume of a cylinder with radius 2 cm and height 5 cm. Use $\pi \approx 3.14$.

14 NUMBER PLANE

Which of these points lie on $y = 3x - 2$: $(1,\ 1)$, $(2,\ 5)$, $(4,\ 10)$?

15 PATTERNS & RULES

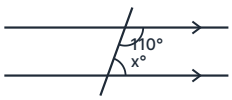
A taxi charges \$4 plus \$2.50 per km. Write a rule for the cost C of a d km trip, then find the cost of a 12 km trip.

16 RATIOS & RATES

A tap fills a 60 L tub in 4 minutes. How long will it take to fill a 225 L tank at the same rate?

17 ANGLES

The lines are parallel. Find x.



18 PROBABILITY

A whole number from 1 to 20 is chosen at random. Find $P(\text{prime})$.

19 DATA

Which number can be added to $3,\ 5,\ 8,\ 10$ so that the mean of the five numbers is 7?

20 PROBLEM SOLVING

A cyclist rides at 18 km/h. How far does she travel in 2 hours 20 minutes?

BONUS

A $3 \times 3 \times 3$ cube is painted red on the outside, then cut into 27 small cubes. How many small cubes have exactly two red faces?

Hints & formulas

Number

Same signs: answer positive. Different signs: answer negative (for × and ÷).

Subtracting a negative is adding: $5 - (-3) = 5 + 3$

$\frac{a}{b} + \frac{c}{d}$: use a common denominator

÷ a fraction: flip it, then multiply

Percent to decimal: ÷ 100. 10% of a number: ÷ 10

★ Increase by 15%: find 15%, then add it on

Indices & roots

$2^4 = 2 \times 2 \times 2 \times 2 = 16$

$\sqrt{49} = 7$ because $7^2 = 49$; $\sqrt[3]{64} = 4$ because $4^3 = 64$

Prime: exactly two factors (1 and itself). 1 is not prime.

★ $a^m \times a^n = a^{m+n}$, $a^m \div a^n = a^{m-n}$

★ Pythagoras: $c^2 = a^2 + b^2$ (c = longest side)

Algebra

Like terms have the same letters: $7x - 3x = 4x$

$3x$ means $3 \times x$; x^2 means $x \times x$

Substitute: replace the letter with the number, use brackets for negatives

$a(b + c) = ab + ac$

★ Factorise: take out the HCF, then check by expanding

Equations

Do the same to both sides

Undo in reverse order: $2x + 5 = 17 \Rightarrow 2x = 12 \Rightarrow x = 6$

Check: substitute your answer back in

★ Unknowns on both sides: collect them on one side first

Measurement

Perimeter: add all the sides. Rectangle $A = lb$

Triangle $A = \frac{1}{2}bh$ · parallelogram $A = bh$

Circle: $C = \pi d = 2\pi r$, $\pi \approx 3.14$

Prism: $V = Ah$ (area of cross-section × length)

$1000\ \text{cm}^3 = 1\ \text{L}$, $1\ \text{km} = 1000\ \text{m}$

★ Circle $A = \pi r^2$ · trapezium $A = \frac{h}{2}(a + b)$ · cylinder $V = \pi r^2 h$

Patterns & graphs

Points are $(x,\ y)$: across, then up

Quadrants: 1st top right, then anticlockwise

A point is on a line if its coordinates fit the rule

Ratios: divide both parts by the HCF. Speed = distance ÷ time

★ Share in a ratio: add the parts, find one part, then multiply

Angles & chance

Straight line $180°$ · point $360°$ · vertically opposite angles are equal

Parallel lines: corresponding and alternate angles are equal; co-interior add to $180°$

Triangle $180°$ · quadrilateral $360°$

$P(E) = \dfrac{\text{favourable outcomes}}{\text{total outcomes}}$, $P(\text{not } E) = 1 - P(E)$

Data & context

Mean = total ÷ number of values

Median: middle value once the data are in order

Mode: most common · range = highest − lowest

Worded problems: write what you know, choose the operation, answer in a sentence with units

Answers

1 Week 1 answers

1. 5 start at -7, move 12 to the right
2. $\frac{3}{5}$ same denominator: add the tops
3. 35% 0.35×100
4. 16 $2 \times 2 \times 2 \times 2$
5. 7 $7^2 = 49$
6. $6a$ $5 + 3 - 2 = 6$
7. 17 $3 \times 4 + 5$
8. $3x + 12$ $3 \times x + 3 \times 4$
9. $x = 6$ subtract 9 from both sides
10. $x = 7$ divide both sides by 4
11. **26 cm** $9 + 4 + 9 + 4$
12. **30 cm²** $\frac{1}{2} \times 10 \times 6$
13. **40 cm³** $5 \times 4 \times 2$
14. **2nd quadrant** x negative, y positive: top left
15. $3,\ 4,\ 5,\ 6$ add 3 to each x value
16. $2 : 3$ divide both parts by 6
17. $a = 50$ angles on a straight line: $180 - 130$
18. $\frac{1}{2}$ $\{2, 4, 6\}$: $\frac{3}{6}$
19. 8 $40 \div 5$
20. **7°C** $-4 + 11$

B **15, 16 and 17** $n + (n + 1) + (n + 2) = 48,\ 3n = 45$

2 Week 2 answers

1. -18 positive × negative = negative
2. $\frac{3}{4}$ divide top and bottom by 6
3. **$15** $\frac{1}{4}$ of 60
4. $2^2 \times 3^2$ $36 = 4 \times 9 = 2 \times 2 \times 3 \times 3$
5. 21 $25 - 4$
6. $10ab$ $2 \times 5 = 10,\ a \times b = ab$
7. 12 $15 - 3$
8. $10y - 15$ $5 \times 2y - 5 \times 3$
9. $x = 11$ add 8 to both sides
10. $x = 6$ $2x = 12$
11. **31.4 cm** $C = \pi d \approx 3.14 \times 10$
12. **40 cm²** $A = bh = 8 \times 5$
13. **27 cm³** $3 \times 3 \times 3$
14. $(5,\ 0)$ on the x-axis, $y = 0$
15. $y = x + 4$ each y is 4 more than x
16. **80 km/h** $240 \div 3$
17. $x = 60$ angle sum of a triangle: $180 - 50 - 70$
18. $\frac{2}{5}$ $\frac{4}{10}$
19. 5 in order: 3, 4, **5**, 7, 9
20. **$365** $170 + 195$

B **4 and 9** $4 \times 9 = 36,\ 4 + 9 = 13$

Answers

3 Week 3 answers

1. 9 — $5 + 4$
2. $\frac{11}{12}$ — $\frac{8}{12} + \frac{3}{12}$
3. 0.24 — $6 \times 4 = 24$; 2 decimal places
4. **23 and 29** — $21 = 3 \times 7$, $27 = 3 \times 9$
5. **5 and 6** — $25 < 30 < 36$
6. $4x + 5y$ — $7x - 3x$ and $4y + y$
7. 11 — $5 - 3(-2) = 5 + 6$
8. $3(2x + 5)$ — HCF = 3
9. $x = 28$ — multiply both sides by 4
10. $x = 5$ — $3x = 15$
11. **9 cm** — $36 \div 4$
12. **58 cm²** — $10 \times 7 - 4 \times 3 = 70 - 12$
13. **30 000 cm³** — $50 \times 30 \times 20$
14. **7 units** — same x: $3 - (-4)$
15. $-1,\ 1,\ 3,\ 5$ — e.g. $2(0) - 1 = -1$
16. **$18 and $27** — 5 parts: $9 each
17. $x = 65$ — corresponding angles are equal
18. 0.7 — $1 - 0.3$
19. **range 8, mode 9** — range $12 - 4$; 9 appears twice
20. **the 2 L bottle** — $1.80 per L vs $2.00 per L

B $41 \times 32 = 1312$ — next best: $42 \times 31 = 1302$

4 Week 4 answers

1. 32 — $20 - (-12) = 20 + 12$
2. $\frac{2}{3}$ $\frac{24}{36}$
3. 0.375 — $3 \div 8$
4. $5^4 = 625$ — 25×25
5. 13 — $4 + 9$
6. $3a$ — $\frac{12ab}{4b}$
7. 7 — $9 + (-2)$
8. $4(2m - 3)$ — HCF = 4
9. $x = 6$ — $15 - 6 = 9$
10. $x = 5$ — $3x + 3 = 18$, $3x = 15$
11. **0.4 km** — $400\ \text{m} \div 1000$
12. **28 cm²** — $\frac{4}{2}(5 + 9) = 2 \times 14$
13. **120 cm³** — $V = Ah = 15 \times 8$
14. $D(1,\ 4)$ — same x as A, same y as C
15. **31** — rule: $3n + 1$; $3(10) + 1$
16. **$22.50** — $2.25 each × 10
17. $x = 145$ — angles at a point: $360 - 120 - 95$
18. $\frac{1}{3}$ — 2 M's out of 6 letters
19. 23 — 9 values; the 5th is 23
20. **138** — 15% of 120 = 18; $120 + 18$

B **50 cm²** — width 5, length 10: $2(5 + 10) = 30$

Answers

5 Week 5 answers

1 4 $-6+10$

2 $4\frac{1}{6}$ $\frac{9}{6}+\frac{16}{6}=\frac{25}{6}$

3 **$12** $10\%=8,\ 5\%=4$

4 3^9 add the indices: $4+5$

5 **10 cm** Pythagoras: $c^2=6^2+8^2=100$

6 $12x^2$ $3\times4=12,\ x\times x=x^2$

7 $P=23$ $15+8$

8 $5x+7$ $2x+10+3x-3$

9 $x=4$ $6x=24$

10 $x=18$ $\frac{x}{3}=6$

11 **31.4 m** $C=2\pi r\approx2\times3.14\times5$

12 **314 cm²** $A=\pi r^2\approx3.14\times100$

13 **2.5 L** $1000\text{ cm}^3=1\text{ L}$

14 **Yes** $2(3)+1=7$ ✓

15 $y=5x+2$ y goes up by 5; starts at 2

16 **160 g** 1 part = 80 g; 2×80

17 $70°$ $(180-40)\div2$

18 $\frac{1}{4}$ $\{7,8\}$: $\frac{2}{8}$

19 **8** total $4\times9=36$; $36-28$

20 **28 m²** $10\times6-8\times4=60-32$

B **60** LCM: $2^2\times3\times5$

6 Week 6 answers

1 31 $16-(-15)=16+15$

2 $\frac{2}{3}$ $\frac{3}{5}\times\frac{10}{9}=\frac{30}{45}$

3 $\frac{2}{5},\ 42\%,\ 0.45$ $0.40,\ 0.42,\ 0.45$

4 7^5 subtract the indices: $8-3$

5 **Yes** $5^2+12^2=25+144=169=13^2$

6 $3x$ $\frac{15}{5}=3,\ \frac{x^2}{x}=x$

7 17 $9+8$

8 $5x(x+2)$ HCF = $5x$

9 $x=4$ $4x-5=11,\ 4x=16$

10 $4n-7=25,\ n=8$ $4n=32$

11 **28 cm** side $=\sqrt{49}=7$; 4×7

12 **55 cm²** $10\times4+\frac{1}{2}\times10\times3=40+15$

13 **62.8 cm³** $V=\pi r^2h\approx3.14\times4\times5$

14 $(1,\ 1)$ **and** $(4,\ 10)$ $3(2)-2=4\neq5$

15 $C=2.5d+4$**; $34** $2.5\times12+4=30+4$

16 **15 minutes** 15 L per minute; $225\div15$

17 $x=70$ co-interior angles add to $180°$: $180-110$

18 $\frac{2}{5}$ 2, 3, 5, 7, 11, 13, 17, 19: $\frac{8}{20}$

19 **9** total must be $5\times7=35$; $35-26$

20 **42 km** $18\times2\frac{1}{3}=36+6$

B **12** one in the middle of each of the 12 edges